

ORIGINAL ARTICLE

The role of peripheral vasopressin 1A and oxytocin receptors on the subcutaneous vasopressin antinociceptive effects

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Abstract

Background: Vasopressin (AVP) seems to play a role as an antinociceptive neurohormone, but little is known about the peripheral site of action of its antinociceptive effects. Moreover, AVP can produce motor impairment that could be confused with behavioural antinociception. Finally, it is not clear which receptor is involved in the peripheral antinociceptive AVP effects.

Methods: In anaesthetized rats with end-tidal CO₂ monitoring, extracellular unitary recordings were performed, measuring the evoked activity mediated by A β -, A δ -, C-fibres and post-discharge. Behavioural nociception and motor impairment were evaluated under subcutaneous AVP (0.1–10 μ g) using formalin and rotarod tests. Selective antagonists to vasopressin (V_{1A}R) or oxytocin receptors (OTR) were used. Additionally, vasopressin and oxytocin receptors were explored immunohistochemically in skin tissues.

Results: Subcutaneous AVP (1 and 10 μ g/paw) induced antinociception and a transitory reduction of the end-tidal CO₂. The neuronal activity associated with A δ - and C-fibre activation was diminished, but no effect was observed on A β -fibres. AVP also reduced paw flinches in the formalin test and a transitory locomotor impairment was also found. The AVP-induced antinociception was blocked by the selective antagonist to V_{1A}R (SR49059) or OTR (L368,899). Immunohistochemical evidence of skin VP and OT receptors is given.

Conclusions: Subcutaneous AVP produces antinociception and behavioural analgesia. Both V_{1A} and OTR participate in those effects. Our findings suggest that antinociception could be produced in a local manner using a novel vasopressin receptor located in cutaneous sensorial fibres. Additionally, subcutaneous AVP also produces important systemic effects such as respiratory and locomotor impairment.

Significance: Our findings support that AVP produces peripheral antinociception and behavioural analgesia in a local manner; nevertheless, systemic effects are also presented. Additionally, this is the first detailed electrophysiological analysis of AVP antinociceptive action after subcutaneous administration. The results are reasonably explained by the demonstration of V_{1A}R and OTR in cutaneous fibres.

1. Introduction

Arginine-vasopressin (AVP), a neurohypophysial hormone well known for its effects on water

homeostasis and blood pressure regulation, has been implied as a potential peptide modulating nociception. In rodents, intracerebroventricular (Kordower

and Bodnar, 1984) and intrathecal (Thurston et al., 1988, 1992; Peng et al., 2015) AVP administration produces behavioural analgesia. Furthermore, a physiological antinociceptive participation of AVP has been suggested by the facts that the AVP-deficient Brattleboro rat strain has hyperalgesia and diminished stress-induced analgesia (Bodnar et al., 1982), and that the systemic hyperosmotic challenge, a model of physiological release of AVP, produces antinociception (Schorscher-Petcu et al., 2010). In this sense, normal AVP plasma concentrations in humans increase from ≈ 4 pg/mL to >13 pg/mL after hypertonic saline infusion (Baylis and Robertson, 1980). Importantly, painful stimuli produce an increment in plasma AVP in rodents (Suzuki et al., 2009) and humans (Kendler et al., 1977; Auböck and Konzett, 1983; Yang et al., 2012), reaching concentrations >25 pg/mL in the latter (Yang et al., 2012).

Systemic AVP (or analogue) administration produces antinociception in chemical, thermal, inflammatory and mechanical nociceptive tests (Berkowitz and Sherman, 1982; Berson et al., 1983; Schorscher-Petcu et al., 2010; Mogil et al., 2011). However, the site of the AVP antinociceptive action after its peripheral administration had not been explored. It has been suggested that AVP could act at dorsal root ganglion (DRG) neurons enhancing GABA_A currents (Qiu et al., 2014a). Recently, Qiu et al. (2014b) showed that subcutaneous AVP diminished the number of flinches induced by acetic acid, suggesting a peripheral local effect. Nevertheless, peripheral AVP administration produces motor impairment (Andrews et al., 1983; Ettenberg et al., 1983), but no information about locomotor effects in behavioural models of pain is available. Furthermore, considering that AVP can cross the blood–brain barrier (Zlokovic et al., 1990), the action site (peripheral vs. central) preceding systemic AVP administration remains ambiguous.

Although the consensus about the receptor involved in AVP-induced antinociception is related to activation of the vasopressin 1a receptor (V_{1A}R) (Schorscher-Petcu et al., 2010; Mogil et al., 2011; Qiu et al., 2014a,b), the role of cutaneous oxytocin receptors (OTR) (González-Hernández et al., 2017) cannot be excluded considering their similitudes (Chini and Manning, 2007). Therefore, the aim of this study was to (1) evaluate the existence of local antinociceptive effects induced by subcutaneous (s.c.) AVP administration, (2) evaluate the participation of V_{1A}R and OTR and (3) differentiate between antinociception and motor impairment. Consequently, we performed electrophysiological

recordings of wide dynamic range (WDR) neurons and used behavioural models of acute pain and locomotor function as well as selective high-affinity V_{1A}R (SR49059) (Serradeil-Le Gal et al., 1993) and OTR (L-368,899) (Williams et al., 1994) antagonists. Using these methods, we found that the s.c. administration of AVP has a local antinociceptive effect that is mainly mediated by V_{1A}R, but we also found OTR participation. In fact, we found V_{1A}R and OTR immunohistochemically positive cutaneous sensorial fibres that could explain our results. Furthermore, AVP produces an important respiratory and motor impairment.

2. Materials and methods

2.1 Experimental animals

Experiments were performed using male Wistar rats ($n = 105$ rats, 280–320 g). Animals were housed in individual cages in a climate-controlled room, with water and food *ad libitum* and a 12-hour light-dark cycle. All experiments were carried out in accordance with the International Association for the Study of Pain (IASP) Guidelines for the Use of Animals in Research (Zimmermann, 1983) and with the approval of the Bioethical Committee at the Instituto de Neurobiología (Bioethical Committee protocols numbers 008, 009 and 010).

2.2 Electrophysiological approach

2.2.1 Surgical procedure

The rats were anaesthetized with urethane (1.2 g/kg; intraperitoneal, i.p.). A tracheotomy was performed to provide artificial ventilation (55 strokes/min, containing two-thirds N₂O and one-third O₂) and to monitor end-tidal CO₂ (Capstar-100 CO₂ analyzer; CWI Inc., Ardmore PA, USA) during the experiment (depicted in Fig. 1A,B,F). The temperature was maintained at 38 °C using a circulating water pad system. The rats were secured in a stereotaxic frame and the lumbar vertebrae were fixed to a spinal frame to improve the stability of the preparation. A laminectomy was performed to expose L4–L6 spinal cord segments, and the dura was carefully removed.

2.2.2 Electrophysiological recordings

Extracellular single unit recordings from deep dorsal horn segments (depth 500–1000 μ m) were made using quartz platinum/tungsten microelectrodes

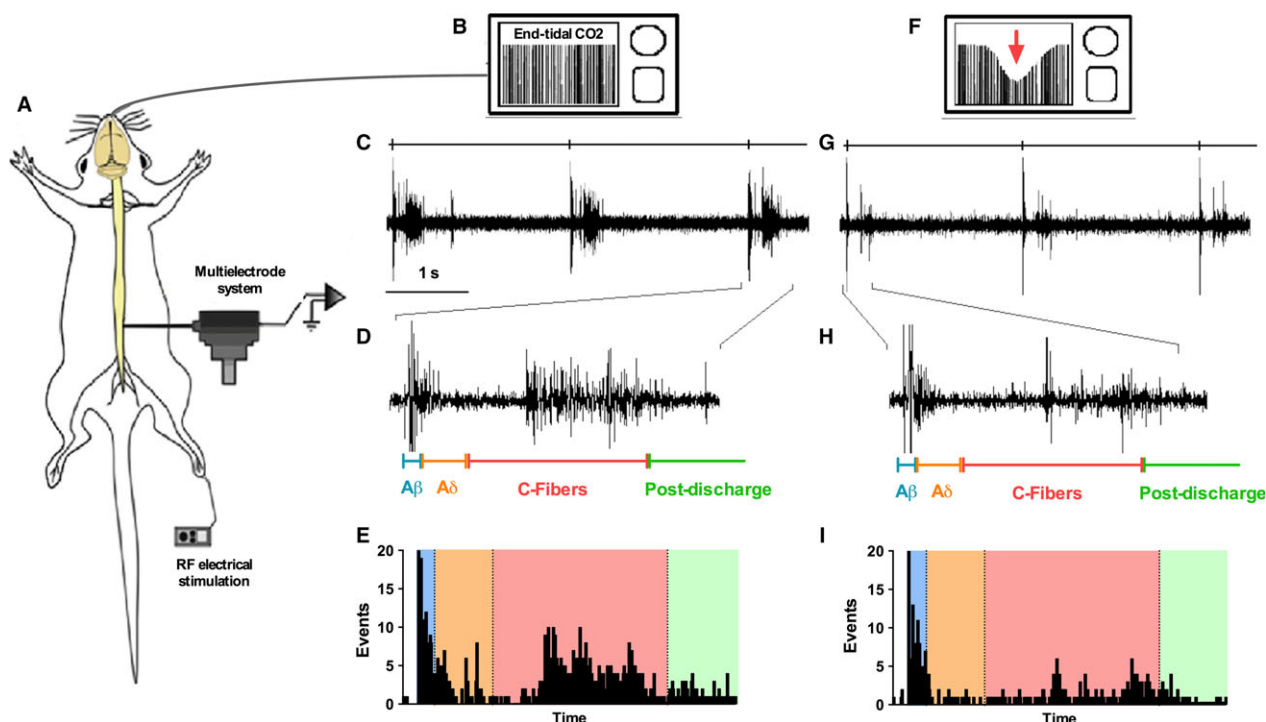


Figure 1 Spinal cord neuronal responses to the receptive field (rf) electrical stimulation. (A) Experimental set-up depicting the recording site at the spinal cord and the rf area of stimulation. (B) Represents the measurement of end-tidal CO_2 with a capnography monitor during the electrophysiological experiments. (C) Stimulus artefact and three responses of WDR spinal neurons after rf electrical stimulation. (D) Single response induced by rf electrical stimulation, and time latencies depicting the different fibre components. (E) Peristimulus time histogram obtained from 20 responses to rf electrical stimulation. (F) Depicts the acute reduction of end-tidal CO_2 10 min after AVP ($10 \mu\text{g}/50 \mu\text{L}$) s.c. administration. (F, G, H, I) show the effect 40 min after AVP ($10 \mu\text{g}/50 \mu\text{L}$) s.c. administration in the rf. Notice the reduction of activity mediated by A δ - and C-fibres, and post-discharge.

($\approx 3\text{--}9 \text{ M}\Omega$). Neuronal activity was recorded, amplified ($\times 1000$), filtered (300 Hz–3 KHz), digitalized and analysed using the Spike 2 system (CED, Cambridge, UK) (Fig. 1A,C,D,G,H). Recorded neurons received afferent input from the ipsilateral hind paw region, and the receptive field (RF) was found with non-noxious tactile stimulation (gentle tapping of the hind paw). Once a single-cell signal responsive to RF stimulation was located, two stainless stimulation electrodes (27 G) were inserted subcutaneously into the RF (Fig. 1A). Stimuli (1-ms pulses at 0.5 Hz; Grass S11 Stimulator) with increasing intensities were given to determine the threshold for activating A δ - and C-fibres (high threshold stimulation). Only cells that responded to both low and high threshold stimulation were used in this study; these cells are classified as wide dynamic range (WDR) neurons. Peristimulus time histograms (PSTH) were used to classify and quantify the evoked WDR neuronal responses according to the following stimulated fibre latencies: A β (0–20 ms), A δ (20–90 ms), C-fibres (90–300 ms) and post-discharge (300–800 ms) (Fig. 1E,I) (Urch and Dickenson, 2003).

2.2.3 Experimental design

Each peristimulus histogram was constructed with a series of 20 electrical pulses (1 ms, 0.5 Hz) at an intensity adjusted to 3 times the threshold required to evoke a C-fibre response. After 4 control series, each one separated by a 5-min interval, AVP (0.1, 1 or $10 \mu\text{g}/50 \mu\text{L}$) was administered s.c. into the RF; then, stimulation was given at minutes 0, 5, 10, 20, 30, 40, 50 and 60 after AVP administration. To evaluate whether the pharmacological effect of AVP was local or systemic, AVP ($10 \mu\text{g}/50 \mu\text{L}$) was administered s.c. into the paw contralateral to the electrical stimulation in one group of rats. To test $V_{1A}R$ or OTR participation in AVP effects, the specific antagonists SR49059 ($10 \mu\text{g}/50 \mu\text{L}$; $V_{1A}R$ antagonist) or L-368,899 ($10 \mu\text{g}/50 \mu\text{L}$; OTR antagonist) were given s.c. in the RF 5 min prior to AVP, and then the aforementioned protocol was followed. As the doses used for the agonist and antagonist could change in their bioavailability and pharmacodynamics due to the subcutaneous administration, the selection followed literature references points (Serradeil-Le Gal

et al., 1993; Williams et al., 1994; Alexander et al., 2015). However, the AVP doses were 0.1, 1 and 10 to have a curve dose response.

2.3 Behavioural tests

2.3.1 Formalin-induced pain behaviour

Animals were habituated to the observation chamber for 30 min for 3 consecutive days prior to the experimental day. On the day of the test, the animals were placed in the observation chamber for 30 min, then AVP (0.1, 1 or 10 µg/50 µL) was injected s.c. into the dorsal surface of the right hind paw; 5 min later, formalin (50 µL of 1% formalin, diluted in saline) was injected s.c. into the same site as the AVP. The nociceptive behaviour (number of paw flinches or shakes) was quantified for 1 min every 5 min over the 60 min following formalin administration; this period was chosen to evaluate the three phases of the formalin test defined as follows: phase 1 (0:10 min), interphase (10:20 min) and phase 2 (20:60 min). To evaluate whether the pharmacological effect of AVP was local or systemic, AVP (10 µg/50 µL) was administered s.c. into the left hind paw (contralateral to the formalin administration) in one group of rats. In the groups with antagonists L-368,899 (100 µg/50 µL) or SR49059 (10 µg/50 µL), the antagonists were given s.c. injections 5 min prior to the AVP at the same site in the hind paw, and then the aforementioned protocol was followed.

2.3.2 Rotarod

A rotarod apparatus (IITC Inc. Life Science, CA, USA) was used to study the administration effects of AVP on locomotor function. The rotarod's velocity was set at 15 revolutions per minute (rpm) with a cut-off time of 180 s (drum diameter 9.52 cm). The rats were trained for 3 consecutive days (2 trainings per day). Training and experimental sessions consisted of 3 consecutive cycles of locomotion. In each cycle, the latency to fall from the rotarod was recorded and then averaged. After the training days, the same group of 9 rats was used for 21 days to evaluate the locomotor effect without treatment (control) and after the administration of saline (vehicle group), AVP (10 µg/50 µL) or OT (100 µg/50 µL). The drugs were injected s.c. into the dorsal surface of the hind paw 10 min prior to the rotarod test, with 24 h between the administration of each treatment.

2.4 Compounds

Apart from the anaesthetic (urethane), the drugs used were [Arg⁸]-vasopressin acetate salt (AVP); oxytocin (OT); (2S)-1-[[[(2R,3S)-5-Chloro-3-(2-chlorophenyl)-1-[(3,4-dimethoxyphenyl)sulfonyl]-2,3-dihydro-3-hydroxy-1H-indol-2-yl]carbonyl]-2-pyrrolidinecarboxamide (SR49059) and (2S)-2-Amino-N-[(1S,2S,4R)-7,7-dimethyl-1-[[[4-(2-methylphenyl)-1-piperazinyl]sulfonyl]methyl]bicyclo[2.2.1]hept-2-yl]-4-(methylsulfonyl)butanamide (L-368,899). All compounds were obtained from Sigma-Aldrich. AVP, OT and L-368,899 were dissolved in saline, whereas SR49059 was dissolved in DMSO (10%).

2.5 Data analysis

2.5.1 Electrophysiology

The mean number of action potentials evoked by peripheral stimuli in the four control series was set as 100% (we only used cells with less than a 10% variation in their activity during the control series). After drug administration, the number of action potentials was quantified in each series (from 0 to 60 min) and normalized compared to the control value. Throughout the experiment, the effect of AVP was evaluated globally as area under the curve (AUC, calculated with the trapezoid method) to unfold the time with the maximum response to AVP. Separate and global statistical comparisons were made between the effect of the different doses of AVP and the reversion by the antagonists. Only one unit was recorded for each animal. *Formalin test.* AUC constructed with the behavioural nociceptive response during phase 1, interphase and phase 2 was calculated. The statistical comparison between the effect of the different doses of AVP and the reversion by antagonists was done separately for each phase. *Rotarod.* The statistical comparison between the mean latencies to fall in each treatment and the control situation was conducted.

Figure data are presented as mean ± SEM (standard error of the mean). In all cases, statistical analysis was performed using GraphPad Prism® V6.01 software (La Jolla, CA, USA). Normal distribution was tested. In the electrophysiological experiments, a two-way repeated measures ANOVA was performed to analyse the effect of the treatments across the course of the experiment. One-way ANOVA was used to compare the global effect of different AVP doses and the participation of V_{1A}R and OTR in peripheral vasopressin effects in the electrophysiological and behavioural

models. Dunnett's post-hoc test was used to compare the treated groups versus the control groups. Tukey's post-hoc test was used when multiple comparisons were assessed. A probability value of $p < 0.05$ was considered statistically significant. The number of animals used in each experiment is reported in the results section.

2.6 Immunofluorescence for OTR and V_{1A}R

Three male Wistar rats (280–310 g) were prepared for an immunofluorescence of skin.

The animals were decapitated and immediately after, the lateral hairy skin areas were excised with a scalpel, snap-frozen in Tissue-Tek O.C.T. Compound (Sakura Finetek USA, Inc, Torrance, CA, USA), placed in liquid nitrogen-chilled isopentane and stored at -80°C before sectioning. A double immunofluorescence for OTR and V_{1A}R was performed on slide-mounted samples of 20- μm thick cryostat sections after fixation in 4% phosphate-buffered paraformaldehyde for 10 min at room temperature. The sections were stored in four consecutive series. The first was used for immunofluorescence as follows: a primary antibody cocktail was used for 24 h at 4°C ; it included polyclonal antibodies against OTR (raised in goat, diluted 1:400, Cat.no. sc-8103; Santa Cruz Biotechnology Inc, Santa Cruz, CA, USA) and V_{1A}R (raised in rabbit, diluted 1:300, Cat.no. A53272; Epigentek Group Inc, USA). The sections were washed for 5 min with 0.1M PBS and then incubated with the appropriate secondary antibody for 24 h at 4°C . The secondary antibodies used were donkey anti-Goat IgG (Alexa Fluor 488 conjugate, Cat.no. A-11055; Invitrogen, USA) and donkey anti-Rabbit IgG (Alexa Fluor 555 conjugate, Cat.no. A-31572; Invitrogen). The second series of sections was treated without the primary antibody as a negative control to avoid a possible false-positive staining. The two remaining series were stored for further analysis.

2.7 Confocal microscopy and image analysis

Confocal images of the lateral hairy skin were acquired using the LSM510 or LSM780 confocal microscope system (Zeiss, Mexico) with $25 \times /0.8$, $40 \times /1.3$ and $63 \times /1.32$ NA oil immersion objective, using the 488-nm argon laser to excite Alexa Fluor 488, and 561-nm diode laser for Alexa Fluor 555. The pinhole, Z-sectioning intervals were kept constant for all images. About 15–30 optical Z-sections of 0.5–1 μm thickness were obtained from the 20- μm thick tissue for each skin image stack.

Optical sections were acquired at a digital size of 1024×1024 pixels and averaged three times to reduce noise. In all cases, the image obtained was improved (brightness and gamma) and analysed using the ZEN 2 Blue Edition Software (Carl Zeiss Microscopy GmbH, Göttingen, Germany). In addition, the colour for each channel was selected as follows: green for OTR and red for V_{1A}R. In all cases, a 2-dimensional projection image and a single optical section image (where the V_{1A}R and OTR signal were strong) were imported in Tag Image File Format (TIFF) and were used to compose the multi-panel figures. In addition, an orthogonal projection using four optical sections was performed.

3. Results

3.1 Electrophysiological approach

3.1.1 Effect of peripheral AVP administration on the afferent spinal cord activity

The electrophysiological recordings of spinal cord neuronal activity showed that the lowest dose of AVP (0.1 $\mu\text{g}/50 \mu\text{L}$, $n = 6$ cells) did not change the neuronal activation evoked by A β -, A δ - and C-fibres nor did it change the post-discharge (Fig. 2). The intermediate AVP dose, 1 $\mu\text{g}/50 \mu\text{L}$ ($n = 5$ cells), did not affect the activity mediated by A β -fibres (Fig. 2A,E,I). Nevertheless, this dose produced a statistically significant reduction in the neuronal activity mediated by A δ -fibres globally (Fig. 2F,J) and a non-significant reduction in the activity mediated by C-fibres (Fig. 2G,K). The peripheral administration of the highest dose of AVP (10 $\mu\text{g}/50 \mu\text{L}$, $n = 6$ cells) did not affect the activity mediated by A β -fibres (Fig. 2A,E,I), but it provoked a reduction in the activity evoked by A δ - (Fig. 2B,F,J) and C-fibres (Fig. 2C,G,K) throughout the experiment. A two-way ANOVA showed a significant reduction of A δ -mediated activity throughout the experiment, except 20 min after AVP administration, and a significant reduction of C-mediated activity starting on minute 20 (Fig. 2B,C) with the highest dose of AVP. In addition, this dose produced a global decrease in post-discharge (Fig. 2H,L) with a maximum effect 30 and 40 min after AVP administration (Fig. 2D). On the other hand, the administration of AVP (10 $\mu\text{g}/50 \mu\text{L}$, $n = 5$ cells) in the paw contralateral to the stimulated paw did not change the neuronal activation evoked by A β -, A δ - and C-fibres (Fig. 2E,F,G,H,I,J,K,L), but a non-significant reduction in post-discharge was observed.

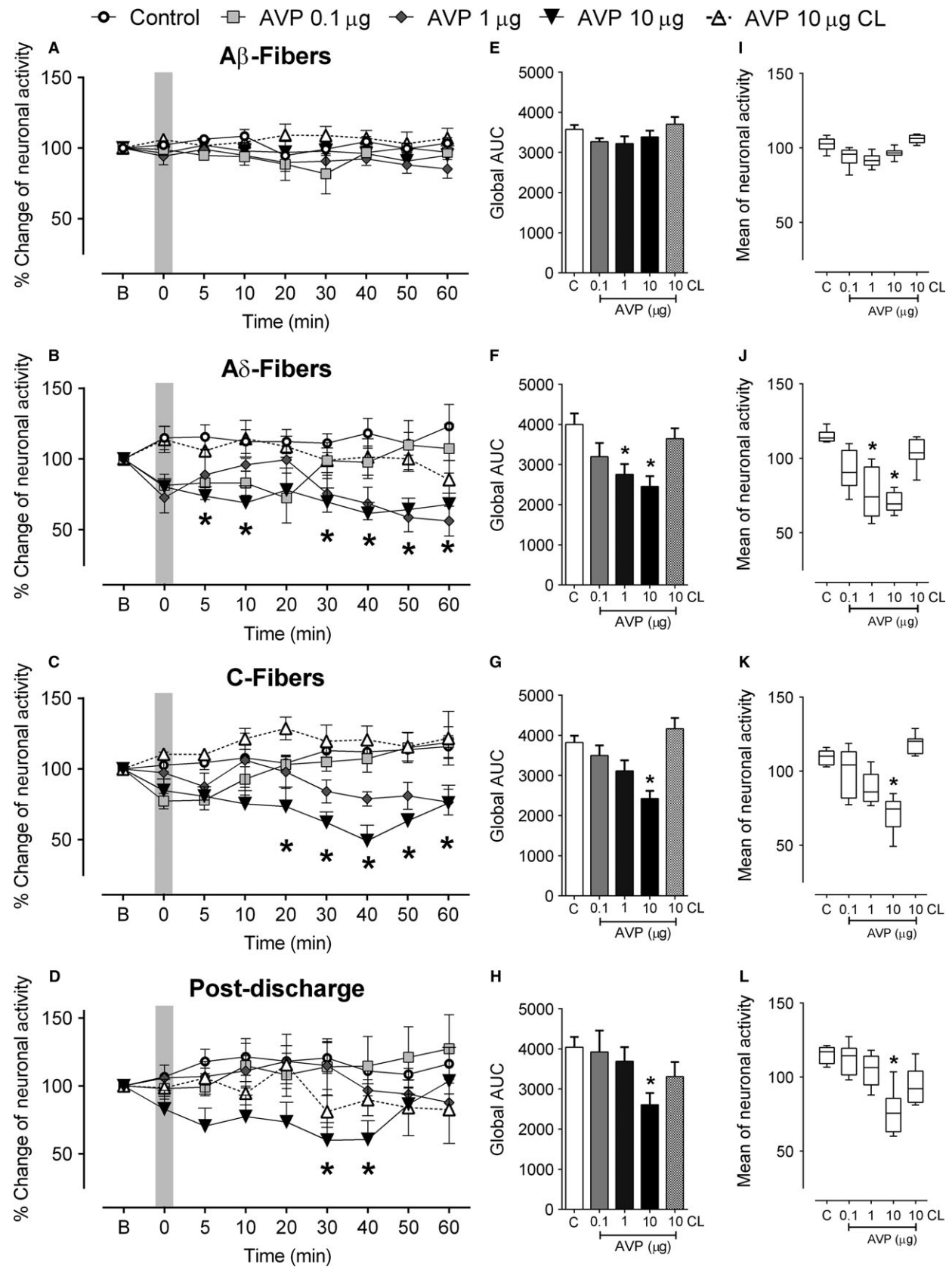


Figure 2 Effect of peripheral AVP administration on the afferent spinal cord activity. (A, B, C, D) graphs show the temporal course of the effects of peripheral AVP administration (dashed line) on the activity of spinal WDR neurons evoked by each type of fibre. * $p < 0.05$ represents the statistical difference between AVP (10 μ g) and the control value at the same time (two-way ANOVA, Tukey's post-hoc test); for the sake of clarity, only the comparison of the maximum AVP dose is presented. (E, F, G, H) show the area under the curve (AUC) of the entire experiment for each treatment. (I, J, K, L) show the global data distribution for each dose. The boxes show the 95th and 5th percentiles, mean, and minimum and maximum values. Notice the inhibition of activity mediated by A δ - and C-fibres, and post-discharge by AVP (10 μ g). * $p < 0.05$, one-way ANOVA, Dunnett's post-hoc test. It is important to notice that AVP in doses of 10 μ g applied in the contralateral (CL) paw produced no significant changes over the afferent activation.

3.1.2 Participation of V_{1A}R and OTR in the effect of peripheral AVP administration on the afferent spinal cord activity

In this set of experiments, we only used the AVP dose, which showed a significant reduction in the activity evoked by A δ - and C-fibres, and the post-discharge from the prior experiments (AVP 10 μ g/50 μ L). Local pretreatment with the specific V_{1A}R antagonist SR49059 (10 μ g/50 μ L, $n = 5$ cells) completely abolished the AVP-induced antinociception throughout the experiment (Figs. 3 and 4). A comparison between the effects of AVP and AVP plus SR

groups revealed a significant difference for A δ , C and post-discharge global activity (Fig. 4F,G,H). Additionally, a two-way ANOVA analysis showed that pretreatment with the V_{1A}R antagonist reverted the effect of AVP in every measurement during the experiment (Fig. 4A,B,C,D). On the other hand, local pretreatment with the specific OTR antagonist, L-368,899 (10 μ g/50 μ L, $n = 5$ cells), produced a partial reversion of the global AVP-induced antinociception. Statistical comparison did not show differences between the AVP plus L3 and control groups nor between the AVP plus L3 and AVP groups when global activity was compared (Fig. 5F,G,H).

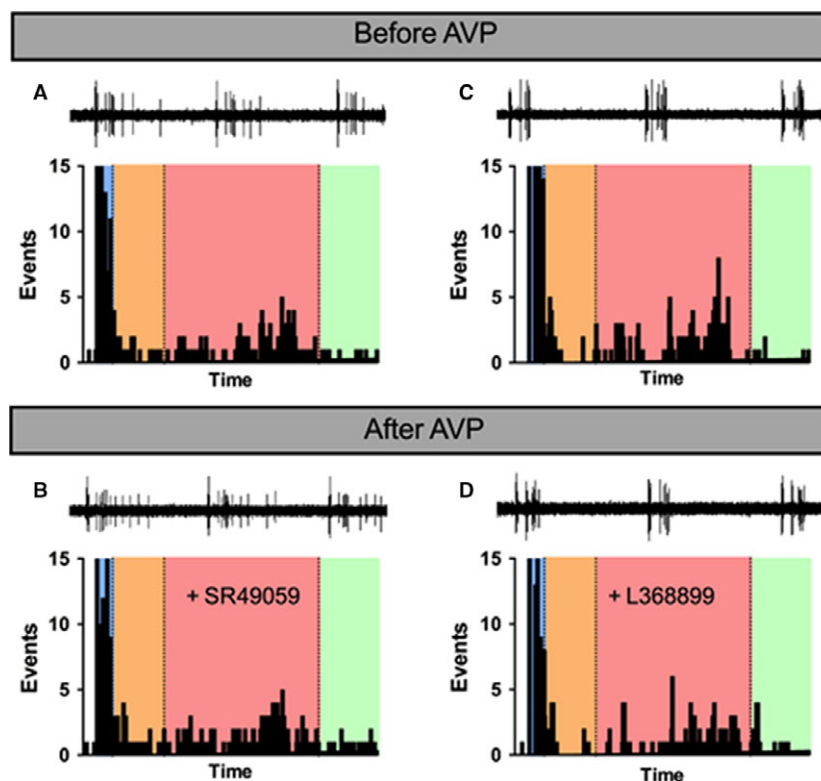


Figure 3 Effect of pretreatment with specific V_{1A}R and OTR antagonists on the AVP actions. (A, B) correspond to the same experiment. (A) Depicts three control responses of a WDR spinal neuron after receptive field (rf) electrical stimulation and a control peristimulus time histogram. (B) Represents the responses after 30 min of AVP administration and pretreatment with V_{1A}R antagonist SR49059. Notice the absence of AVP-induced reduction in A δ - and C-fibre-evoked activity. (C, D) correspond to the same experiment. (C) Depicts control responses and peristimulus time histogram. (D) Represents the responses after 30 min of AVP administration and pretreatment with OTR antagonist L368899. Notice the blockade of AVP reduction in A δ - and C-fibre-evoked activity.

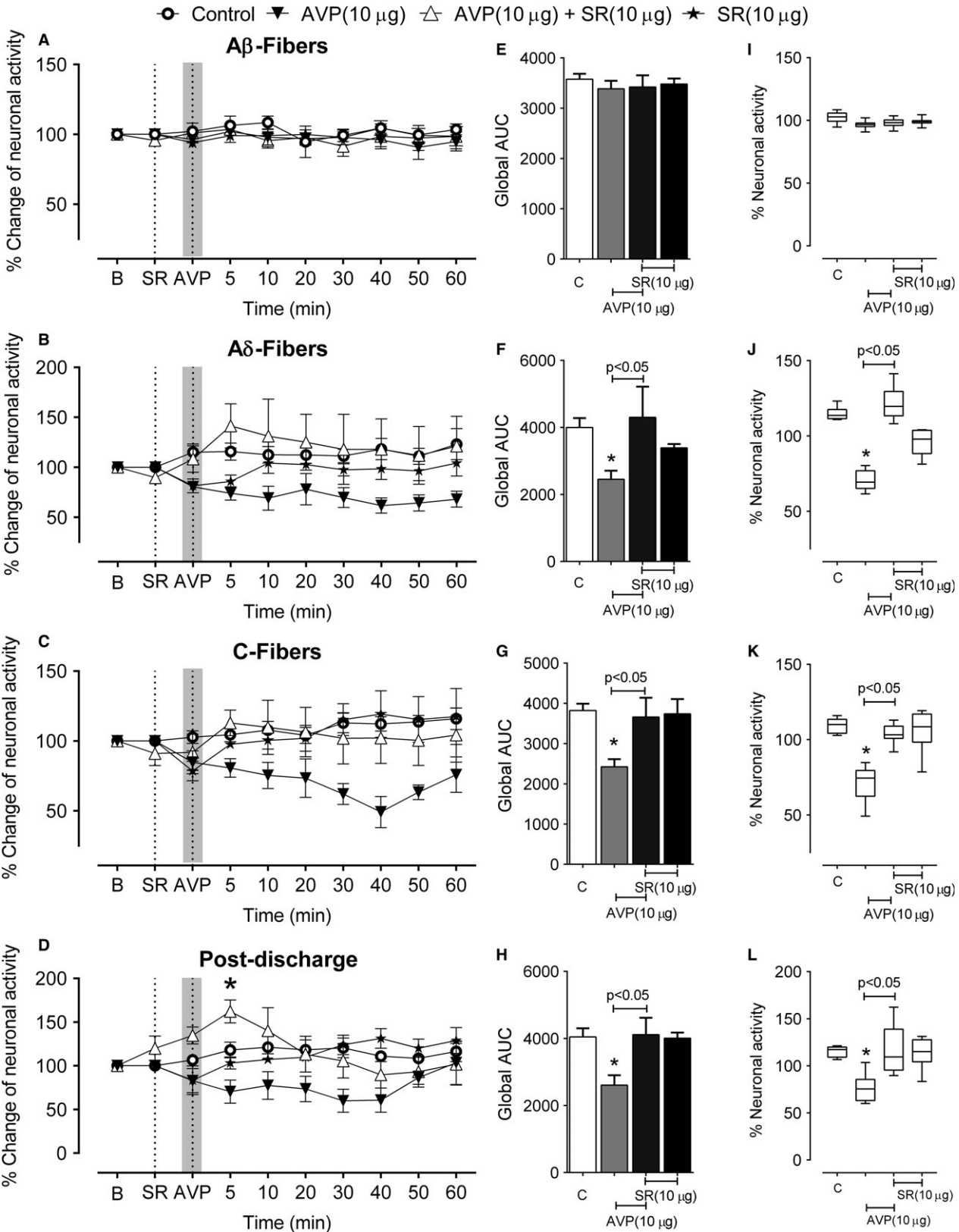


Figure 4 V1a receptor participation in the effects of peripheral AVP administration. (A, B, C, D) graphs show the temporal course of the effects of peripheral AVP (dashed line in darkened area) in animals pretreated with the V_{1A}R antagonist (SR49059) (dashed line). **p* < 0.05 represents the statistical difference between AVP (10 µg) + V_{1A}R antagonist (10 µg) compared with the control value at the same time (two-way ANOVA, Tukey's post-hoc test); for the sake of clarity, only the comparison of the AVP + V_{1A}R antagonist is presented. (E, F, G, H) show the area under the curve (AUC) of the entire experiment for each treatment. (I, J, K, L) show the global data distribution for each dose. The boxes show the 95th and 5th percentiles, mean, and minimum and maximum values. Notice the blockade produced by V_{1A}R antagonist over the response produced by peripheral AVP (10 µg). **p* < 0.05, one-way ANOVA, Tukey's post-hoc test.

Nevertheless, a two-way ANOVA analysis showed that pretreatment with OTR antagonist prevented the effect of AVP throughout the experiment, except for Aδ activity 60 min after AVP administration (Fig. 5A,B,C,D). Neither V_{1A}R antagonist nor OTR antagonist administration produced an effect per se on Aβ-, Aδ-, and C-fibres, and post-discharge evoked responses (*n* = 5, *n* = 6 cells per group, respectively) (Figs. 4 and 5).

3.2 Effect of peripheral AVP administration on end-tidal CO₂

As described in the Surgical Procedure section, end-tidal CO₂ was recorded during the electrophysiological experiments. After s.c. administration of AVP (1 and 10 µg/50 µL, *n* = 6 rats per group), we found a reduction in end-tidal CO₂ between minutes 5 and 20. This effect was not present after OT administration (31 µg/50 µL, *n* = 6 rats) (Fig. 6). S.c. preadministration of V_{1A}R (10 µg/50 µL, *n* = 5) and OTR (10 µg/50 µL, *n* = 6) antagonists did not prevent the end-tidal CO₂ reduction produced by AVP (data not shown).

3.3 Behavioural approach

3.3.1 Effect of peripheral AVP administration on formalin-induced pain behaviour

Peripheral administration of AVP (0.1 µg/50 µL, *n* = 6) failed to modify any of the three formalin test phases (Fig. 7A,B). AVP (1 µg/50 µL, *n* = 6) reduced the behavioural nociceptive response in phases 1 and 2 of the formalin test (Fig. 7A,B). AVP (10 µg/50 µL, *n* = 6) diminished the behavioural nociceptive response in all phases of the formalin test (phase 1, interphase and phase 2) (Fig. 7A,B). In contrast, the contralateral administration of AVP (10 µg/50 µL, *n* = 5) only diminished the behavioural nociceptive response in phase 1 and the interphase (Fig. 7A,B).

3.3.2 Participation of V_{1A}R and OTR in the effect of AVP in the formalin test

Local pretreatment with either the V_{1A}R antagonist SR49059 (10 µg/50 µL) or the OTR antagonist

L-368,899 (100 µg/50 µL) failed to modify the effect of AVP (10 µg/50 µL) in phase 1 and the interphase; however, both antagonists partially reverted phase 2 of the formalin test (*n* = 6 per group) (Fig. 7C,D). The sole administration of SR49059 (10 µg/50 µL, *n* = 7) provoked a statistically significant increment in the behavioural nociceptive responses during the interphase and phase 2 of the formalin test (Fig. 7C). The antagonist L-368,899 had no significant effects (Fig. 7D).

3.3.3 Effect of peripheral AVP and OT administration on locomotor function

The s.c. administration of AVP (10 µg/50 µL, *n* = 10 rats) produced locomotor impairment as shown by the decrease in latency to fall. Locomotor impairment was totally absent 1 h after AVP administration. In contrast, the administration of OT (100 µg/50 µL) did not produce motor impairment (Fig. 8).

3.4 V_{1A}R and OTR immunofluorescence

A double immunofluorescence technique against V_{1A}R and OTR was accomplished in the lateral hairy skin tissues. This technique allowed us to observe in the same tissue of both types of receptors. The presence of V_{1A}R was clearly detected in thin cutaneous fibres, and we confirmed the previous results of OTR fibres in the skin (González-Hernández et al., 2017). We tried to verify V_{1A}R and OTR co-localization using the orthogonal view tool since both receptors were very close to each other in several regions and samples of both receptors. However, we did not have any evidence of co-localization between the two receptors. Figure 9 illustrates an example of at least two parallel fibres, one with V_{1A}R and another with OTR.

4. Discussion

Taken together, these results show that s.c. AVP administration produces two discernible effects: (1) antinociception and behavioural analgesia; and (2) systemic responses (respiratory and locomotor impairment) that seem unrelated to antinociception but could temporarily predominate over the former.

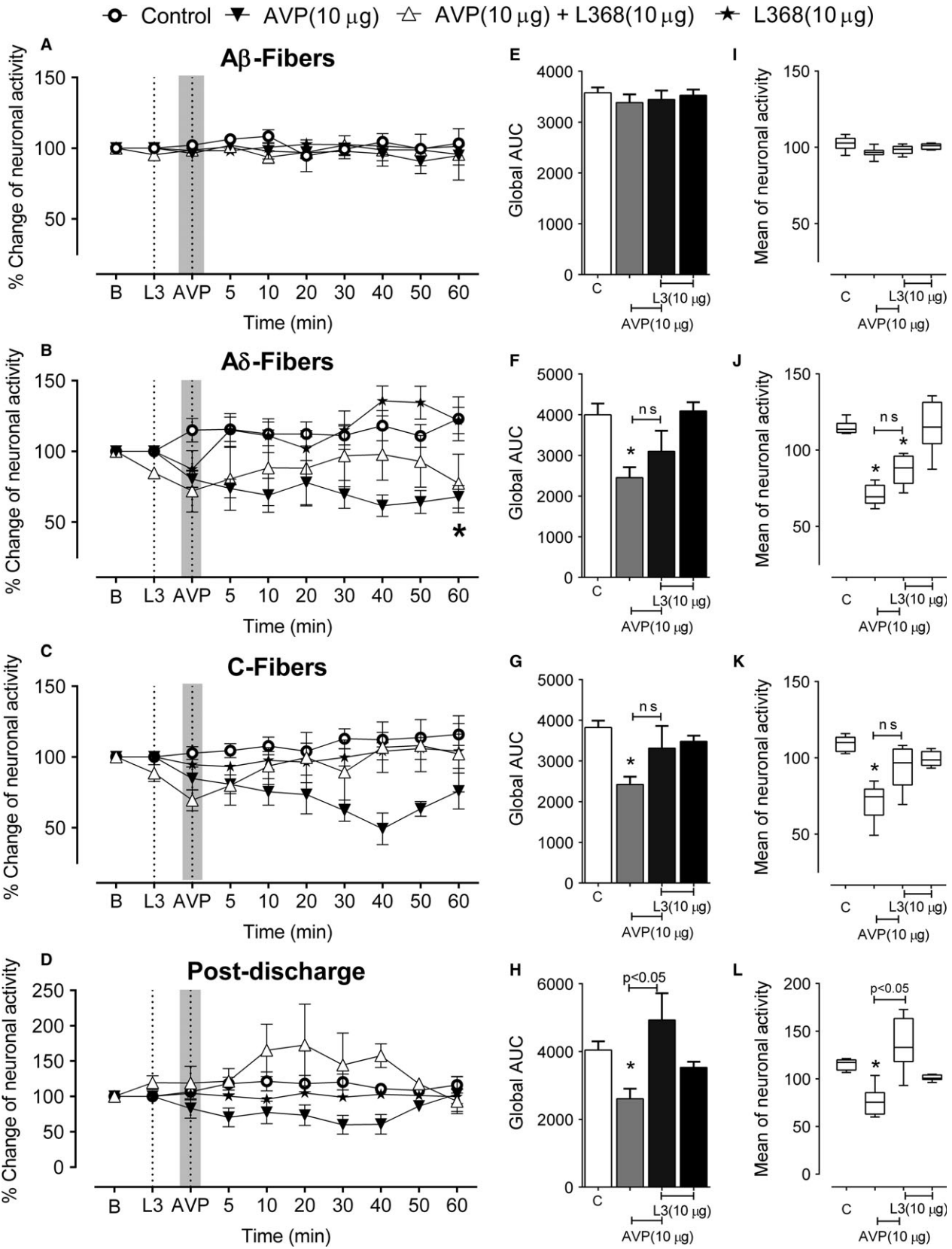


Figure 5 OT receptor participation in the effects of peripheral AVP administration. (A, B, C, D) Graphs showing the temporal course of the effects of peripheral AVP (dashed line in darkened area) in animals pretreated with the OTR antagonist (L368,899) (dashed line). Statistical difference between AVP (10 μ g) + OTR antagonist (10 μ g) compared with the control value at the same time (two-way ANOVA, Tukey's post-hoc test) was assessed. (E, F, G, H) show the area under the curve (AUC) of the entire experiment for each treatment. (I, J, K, L) show the global data distribution for each dose. The boxes show the 95th and 5th percentiles, mean, and minimum and maximum values. Notice the partial blockade produced by OTR antagonist over the response produced by peripheral AVP (10 μ g) (F, J, G, K). * $p < 0.05$, one-way ANOVA, Tukey's post-hoc test.

Finally, our findings could be mediated by the presence of $V_{1A}R$ and OTR in cutaneous fibres.

4.1 Antinociception induced by peripheral AVP administration

AVP as an antinociceptive molecule has been previously addressed (Berkowitz and Sherman, 1982; Berson et al., 1983; Schorscher-Petcu et al., 2010; Mogil et al., 2011; Qiu et al., 2014a,b). It was hypothesized that systemic AVP could produce peripheral antinociception by up-regulating the function of the GABA_A receptor (Qiu et al., 2014a) or inhibiting acid-sensing ion channels in the DRG (Qiu et al., 2014b). To our knowledge, only one work previously addressed the possible peripheral local effect of AVP administration (Qiu et al., 2014b). Qiu et al. (2014b) described that OT and AVP produced local antinociception after intraplantar injection of acetic acid. In fact, using electrophysiological and behavioural approaches, our results show that sub-cutaneous AVP produces antinociception.

It is well accepted that A δ - and C-fibres of primary afferent carry the nociceptive information to the spinal cord, whereas A β -fibres preferentially transmit non-nociceptive information (Basbaum et al., 2009). Thus, the finding that s.c. AVP in the receptive field reduces the activity mediated by A δ - and C-fibres, but not by A β -fibres, is interpreted as a specific antinociceptive effect. However, it is

important to remark that only the s.c. administration in the receptive field had this effect, whereas the s.c. administration in the paw contralateral to the stimulated paw did not diminish the activity evoked by A δ - and C-fibres.

Subsequently, we studied whether this antinociceptive effect could transform into behavioural analgesia using the formalin test. Formalin administration produces paw shakes with three distinguishable phases (Barrot, 2012). Phase 1 (0–10 min), characterized by intense paw shakes, is possibly mediated by the activation of TRP channels (McNamara et al., 2007). The nociceptive behaviour diminishes during the interphase (10–20 min), but it is not known if this decrease is mediated by central modulation or by the absence of formalin activity (Matthies and Franklin, 1992; Fischer et al., 2014). Finally, phase 2 (20–60 min), characterized by the reappearance of nociceptive behaviour, is probably induced by peripheral inflammation and central sensitization (Hunskar and Hole, 1987). Considering that both ipsilateral (to the site of formalin application) and contralateral AVP administration (10 μ g/50 μ L) diminished the response in phase 1 and interphase, this effect could be correlated with the locomotor impairment observed and described elsewhere (Andrews et al., 1983; Ettenberg et al., 1983). However, only ipsilateral AVP administration produced an important reduction in nociceptive behaviour during phase 2 of the formalin test. In fact, our electrophysiological data show an early reduction of activity evoked by A δ -fibres starting 5 min after AVP administration followed by a reduction in C-fibres and post-discharge when applied only in the same paw where formalin was administered, so it is possible that an early local effect could be masked by the systemic response.

Additionally, it remains to be addressed whether AVP could participate in other mechanisms related to nociception. AVP has multiple anti-inflammatory effects (Russel and Walley, 2010) that could contribute to the reduction in the phase 2 of the formalin test described in this study. It is interesting to notice that $V_{1A}R$ knockout mice present an increased response in phase 2 of the formalin test (Schorscher-Petcu et al., 2010), which is consistent with our data when the $V_{1A}R$ antagonist alone incremented this phase.

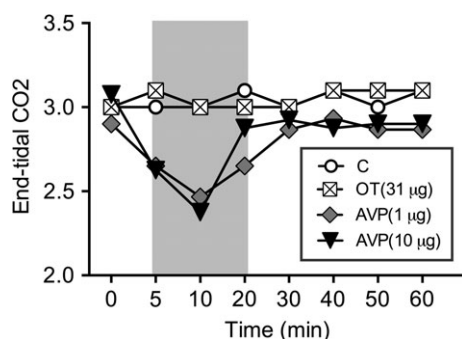


Figure 6 Effect of peripheral AVP administration on the end-tidal CO₂. The graph shows the time course of end-tidal CO₂ after peripheral AVP administration. The grey area indicates the maximum decrease in end-tidal CO₂ after 1 and 10 μ g AVP administration. Notice that OT does not produce this effect.

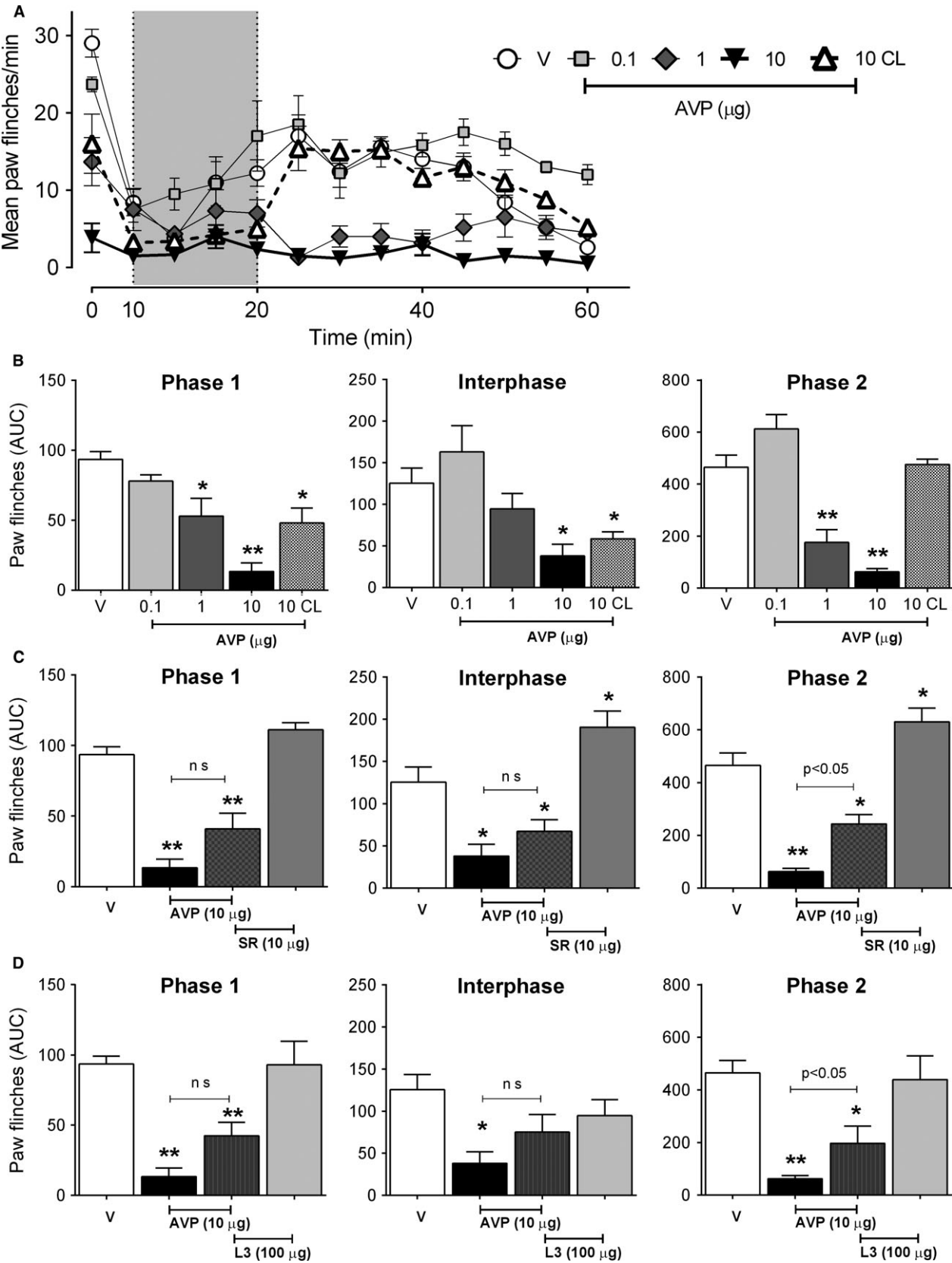


Figure 7 Effect of peripheral AVP on the formalin nociceptive-evoked response. (A) Time course of formalin paw flinches after the administration of AVP at different doses. The grey area represents the interphase between the two formalin-evoked responses. Notice the absence of flinch responses produced by ipsilateral 10 μ g compared with the contralateral 10 μ g of AVP during phase 2 of the formalin test. (B) Area under the curve (AUC) quantification of AVP effect on the formalin response throughout the different phases: phase 1 (0–10 min), interphase (10–20 min) and phase 2 (20–60 min). (C) AUC analysis showed the participation of V1a receptor in the effect of peripheral AVP in the formalin test. (D) Shows the participation of OT receptor in the effect of peripheral AVP in the formalin test. * $p < 0.05$, ** $p < 0.01$ one-way ANOVA, Dunnett's post-hoc test for B, Tukey's post-hoc test for C and D.

4.2 Participation and location of V_{1A}R and OTR in the antinociceptive effects of peripheral AVP

AVP (Peng et al., 2015) and OT produce central (Condés-Lara et al., 2008; DeLaTorre et al., 2009) and peripheral (Hobo et al., 2012; Qiu et al., 2014b) antinociception; however, the receptor involved in those effects is still a matter of debate (González-Hernández et al., 2014). In this regard, administration of AVP or OT in V_{1A}R knockout mice failed to evoke behavioural analgesia (Schorscher-Petcu et al., 2010; Mogil et al., 2011; Qiu et al., 2014b). Nevertheless, there is evidence of OTR participation in central (Miranda-Cardenas et al., 2006) and peripheral (Juif and Poisbeau, 2013; Gong et al., 2015) antinociception. We found that both receptors seem to be implicated in peripheral AVP antinociception. Owing to the highly similar structure of AVP and OT, the former could bind with similar affinity to V1a and OT receptors in *in vitro* experiments (Manning et al., 2008). Consequently, we used two high-affinity, non-peptide antagonists to study the participation of both receptors (Serradeil-Le Gal et al., 1993; Williams et al., 1994). In our electrophysiological approach, the AVP-induced antinociception appears to be preferentially mediated by V_{1A}R although OTR also participates. Similarly, both antagonists partially blockade the antinociceptive effect of AVP in phase 2 of the formalin test. It is possible that a complete reversion did not occur with either antagonist because of the multiple nociceptive

factors involved in this behavioural test; for example, the numerous peripheral inflammatory mediators (Shibata et al., 1989) and central sensitization mechanisms (Tjølsen et al., 1992). Our immunohistochemical results demonstrated that peripheral OTR and V_{1A}R are present and that AVP could participate using both receptors due to the lack of total reversion with the specific antagonists. Nevertheless, we never used both antagonists simultaneously.

In DRG, V_{1A}R mRNA is markedly expressed in small and medium-diameter cells (Schorscher-Petcu et al., 2010), whereas OTR has been found co-localizing with both IB4 and CGRP (Moreno-López et al., 2013; Gong et al., 2015). OTR and V_{1A}R seem to be located in very close proximity, but we could not specify if they co-localized or only co-existed. Our results confirmed the existence of OTR in peptidergic peripheral nerve terminals (González-Hernández et al., 2017) and revealed the presence of V_{1A}R in peripheral fibres.

4.3 Systemic versus local AVP effects

Systemic effects of AVP considered the transitory diminution of end-tidal CO₂ and the locomotor impairment.

4.3.1 End-tidal CO₂

Systemic AVP produces vasoconstriction, preferentially in non-vital organs (László et al., 1991). Our

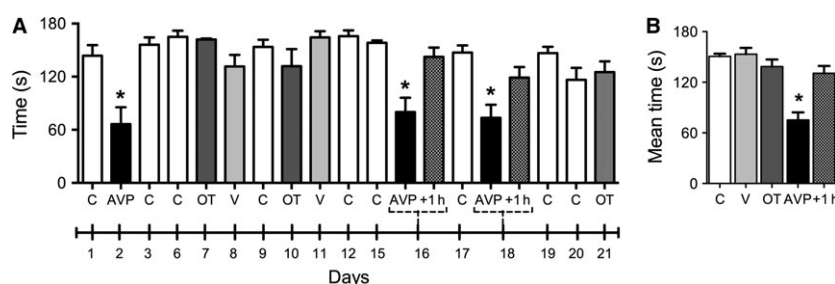


Figure 8 Locomotor effects induced by peripheral AVP administration. (A, B) Shows the effects on a motor function studied with the rotarod test. (A) Course over 21 days of the motor effects induced by AVP administration (10 μ g), OT (100 μ g), vehicle (V-saline) or control (C) measurements. AVP+1 h represents the effect on motor function 1 h after AVP administration. In (B), the mean time of the different treatments along the days is presented. * $p < 0.05$, one-way ANOVA, Dunnett's post-hoc test.

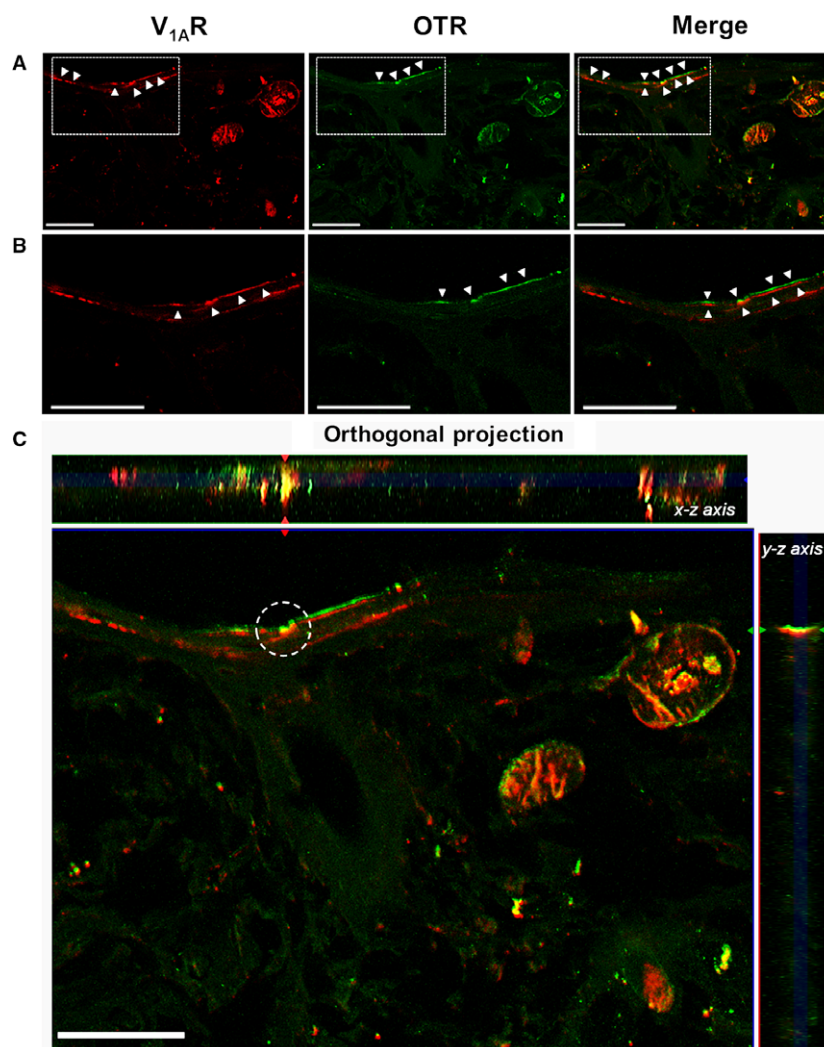


Figure 9 Representative confocal image of a skin (dermis and epidermis) section for $V_{1A}R$ and OTR immunofluorescence. The images are double-labelled sections of dermis and epidermis with antibodies against $V_{1A}R$ (red) and OTR (green) in the lateral hairy skin paw. The arrows show the site where $V_{1A}R$ and OTR are preferentially present in the dermis/epidermis. Panels in A are the z-stack projection from 4 optical sections ($25\times/0.8$ NA objectives). Panels in B are a single optical section cropped from panels A (from the box) showing that $V_{1A}R$ and OTR signals are near but not co-localized; the optical section was selected at the point with the best signal for $V_{1A}R$ and OTR. Panel C shows an orthogonal view displaying four optical sections at $25\times/1.3$ NA; y-z plane and x-z plane show no co-localization between $V_{1A}R$ and OTR. Scale bars: 50 μm .

finding of end-tidal CO_2 reduction could be explained by a transitory hypoperfusion in non-vital vascular beds and a subsequent diminished transport of CO_2 from the periphery to the lungs (Anderson and Breen, 2000). On this subject, although the relation of a hypertensive response to AVP and antinociception was not approached in the current study, previous studies have explored this possibility and have not found a correlation between pressor and analgesic effects of AVP (Berkowitz and Sherman, 1982; Berson et al., 1983).

4.3.2 Locomotor impairment

It has been reported that the systemic AVP can produce locomotor impairment (Andrews et al., 1983; Ettenberg et al., 1983) and this effect can be confused with antinociception, when nociceptive behaviour is estimated as motor responses (Cartmell et al., 1991). We confirmed the AVP induced locomotor impairment with the rotarod test. This transitory effect could have a role in the diminish of phase 1 and interphase of the formalin test by both ipsi and contralateral AVP administration. Nonetheless,

during phase 2 only ipsilateral AVP produced behavioural antinociception, suggesting that this is a local effect unrelated to locomotor function.

4.3.3 Central versus peripheral AVP antinociceptive effects

AVP has multiple antinociceptive actions in the central nervous system. Microinjection of AVP in the nucleus raphe magnus (Yang et al., 2006b), caudate nucleus (Yang et al., 2006a) and spinal cord (Peng et al., 2015) produces antinociception. The participation of opiate and serotonin systems in AVP central antinociception has also been reported (Yang et al., 2009). Additionally, it is possible that circulating AVP could cross the blood–brain barrier (Zlokovic et al., 1990). Although the participation of central effects after peripheral AVP administration could not be explored in this study, it is unlikely that they play a major role in our results on the antinociceptive effect. Indeed, Rojas-Piloni et al. (2010) used a similar electrophysiological model and reported that spinal AVP microinjection did not diminish the responses evoked by A δ - and C-fibres. On the other hand, in the current study, the contralateral (to electrical stimuli and to formalin) administration of AVP neither diminished the activity mediated by A δ - and C-fibres nor produced the behavioural antinociceptive effect during phase 2 of the formalin test. Consequently, the presence of a V_{1A}R in the periphery seems to be responsible for the AVP action.

5. Conclusion

Subcutaneous AVP diminished the evoked activity mediated by A δ - and C-fibres and produced behavioural antinociception with V_{1A}R and OTR implications. The immunohistochemical location of V_{1A}R and OTR in cutaneous fibres suggests their mediation in the local effects. However, subcutaneous AVP administration produces respiratory CO₂ decrement and locomotor impairment that could be confused with antinociception.

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Author contributions

A.M.G. contributed to the acquisition, analysis and interpretation of data and participated in drafting the manuscript. A.G.H. contributed to the conception and design of the study; to the acquisition, analysis and interpretation of data; and participated in drafting the manuscript. I.A.T.G. contributed to the acquisition, analysis and interpretation of data. G.M.L. contributed to the design of the study and to the analysis and interpretation of data. M.C.L. guided the research group, contributed to both the conception and the design of the study; to the analysis and interpretation of data; and participated in drafting and finalizing the manuscript. All authors participated in a critical review of the final manuscript.

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